

Instructions. This homework involves writing dynamic SQL code and a write-up. Use the “CIA World Factbook” database you created in HW-3 to implement the following program. Note that you can use Python, Java, or C++ for the assignment.¹ The program you write should act as a simple text-based interface over your database as described below. Note that your program can assume valid values are given except where noted below (the focus here is on practicing using SQL within a programming language, not on your ability to write a text-based UI). *For full credit, your program file must contain file comment headers with your name, the class, the year, and a brief description (at a minimum). Additionally, as described in class, you must use the appropriate library calls for all user input included in queries (to avoid SQL injection attacks). If at any point your program does not do this, you will get an automatic 0 on the assignment.* Be sure to place your code in the “root” directory of your GitHub repo. Additional instructions on obtaining the starter code and submitting your assignment will be provided through piazza.

1. After starting your program, it should display the following main menu.

```

1. List countries
2. Add country
3. Add border
4. Find countries based on gdp and inflation
5. Update country's gdp and inflation
6. Remove border
7. Exit
Enter your choice (1-7):

```

2. If a user selects 1 from the main menu, your program should display the names, codes, gdp, and inflation of all of the countries. Each country should be displayed on a single line as:

```

name (code), per capita gdp $gdp, inflation rate inflation%

```

For example, according to the CIA World Fact Book (for 2023), the US would be printed as:

```

United States (US), per capita gdp $73600, inflation rate 4.12%

```

After listing all of the countries, your program should re-prompt the main menu.

3. If a user selects 2 from the main menu, your program should prompt for the country information to add to the database.

```

Country code.....: CA
Country name.....: Canada
Country per capita gdp (USD)...: 56900
Country inflation (pct).....: 6.8

```

¹Examples of programs in each language will be provided in Piazza.

Once given, your program should (a) check to make sure the same country code does not already exist, and if not, (b) add the corresponding country to the database, and otherwise, (c) notify the user that the country already exists. Once completed, your program should reprompt the main menu.

4. If a user selects 3 from the main menu, your program should prompt for the border information to add to the database.

```
Country code 1..: CA
Country code 2..: US
Border length...: 8891
```

Once given, your program should (a) check to make sure the border doesn't already exist (note: the border exists if CA is country 1 and US is country 2 or US is country 1 and CA is country 2), and if not, (b) add the corresponding border to the database, and otherwise, (c) notify the user that the border already exists. Once completed, your program should reprompt the main menu.

5. If a user selects 4 from the main menu, your program should prompt for the following information.

```
Minimum per capita gdp (USD)...: 10000
Maximum per capita gdp (USD)...: 80000
Minimum inflation (pct).....: 1.5
Maximum inflation (pct).....: 12.5
```

Once given, your program should then display all countries within the given ranges (inclusive). The countries should be displayed from highest-to-lowest gdp such that if two countries have the same gdp, they should be displayed from lowest-to-highest inflation. (Note you may need to add additional countries to your database to demonstrate that "ties" are working correctly.) Each matching country should be displayed on a single line formatted similarly to above. Note that your *program* should not perform any sorting, and instead all sorting *must* be performed by the DBMS (via SQL). After the corresponding countries are displayed, your program should reprompt the main menu.

6. If a user selects 5 from the main menu, your program should prompt for the country code, new gdp, and new inflation.

```
Country code.....: CA
Country per capita gdp (USD)...: 55800
Country inflation (pct).....: 3.88
```

Once given, your program should (a) check to make sure the country code already exists, and if so, (b) update the corresponding gdp and inflation, and otherwise, (c) notify the user that the country does not exist. Once completed, your program should reprompt the main menu.

7. If a user selects 6, your program should prompt for the border to remove from the database.

Country code 1..: US

Country code 2..: CA

Once given, your program should (a) check to make sure the border exists (again, it could exist in either “direction”), and if it does, (b) remove the corresponding border from the database, and otherwise, (c) notify the user that the border does not exist. Once completed, your program should reprompt the main menu.

8. If a user selects 7, your program should halt.

Submission: *Please read the following carefully and follow all instructions below.* Submit your source code file to canvas for the assignment. Your program file should be named **hw5.py**, **hw5.cpp**, or **HW5.java** if you are using Python, C++, or Java, respectively. Your code must be reasonably modularized (i.e., broken into basic functions as opposed to one large function), commented, correctly handle user input to avoid SQL injection attacks, and be consistently and properly formatted. You *must* also submit a brief write-up of your assignment to canvas as a single PDF file called **hw-5.pdf** that: (1) shows that your code works correctly (e.g., via screen shots) and (2) describes any challenges you had and how you addressed them. For (1), you must include screen shots for each menu selection displaying user input, the corresponding output, and the update to the database (unless no update is made to the database). You must also include screenshots for the cases where countries are selected by the user that are not present in the database (e.g., for menu items 3, 5, and 6) as well as the case where the country already exists for menu item 2. Your write-up must be organized and easy for the grader to verify that the program met all of the requirements listed above. Finally, you must submit a **hw5.sql** file with your CIA World Factbook tables (i.e., DROP TABLE, CREATE TABLE, and INSERT INTO statements). The grader must be able to use your **hw5.sql** file to create the database and your program file to run your program.